Homework 3

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```
x <- sample(c(-10:10, rep(NA, 5)), sample(20:40, 1), replace = TRUE)
x
```

```
##  [1]  -5  -5   3  -6  NA  10   5  -8  NA -10   2   9  -1   7  -7   3   0  -6  -8
## [20]   3  NA   8  NA
```

1. Find and display the last element of the vector.

```
tail(x, n = 1)
```

```
## [1] NA
```

2. Display every other element of the vector, that is, the ones in odd indices.

```
x[c(seq(1, length(x), by = 2))]
```

```
##  [1] -5  3 NA  5 NA  2 -1 -7  0 -8 NA NA
```

3. Display the non-missing values, that is, those that are not NA

```
x[!is.na(x)]
```

```
##  [1]  -5  -5   3  -6  10   5  -8 -10   2   9  -1   7  -7   3   0  -6  -8   3   8
```

4. Display sum of squared values when NA values have been removed

```r
sum(x[!is.na(x)]^2)
```

```
## [1] 750
```

5. Display the number of unique entries that are in the vector.

```r
unique(x)
```

```
##  [1]  -5   3  -6  NA  10   5  -8 -10   2   9  -1   7  -7   0   8
```

6. (Grad student only) Change the sign of any values of the vector that are multiples of three.

```r
x_sign_changed<-ifelse(!is.na(x) & x %% 3 == 0, -x,x)
x_sign_changed
```

```
##  [1]  -5  -5  -3   6  NA  10   5  -8  NA -10   2  -9  -1   7  -7  -3   0   6  -8
## [20]  -3  NA   8  NA
```